$$= \lim_{n\to\infty}\left[15+\frac{54}{n^3}\frac{n(n+1)(2n+1)}{6}+\frac{36}{n^2}\frac{n(n+1)}{2}\right]$$

$$= \lim_{n\to\infty}\left[15+9\left(1+\frac{1}{n}\right)\left(2+\frac{1}{n}\right)+18\left(1+\frac{1}{n}\right)\right]$$

$$= 15+9(1+0)(2+0)+18(1+0) = 51.$$

EXERCISE 9.2

1. Evaluate the following integrals as the limits of sums:

(i) $\int_0^1 (5x+4)dx$ (ii) $\int_1^2 (4x^2-1)dx$

9.3 Fundamental Theorems of Integral Calculus and their Applications

We observe in the above examples that evaluation of $\int_a^b f(x)dx$ as a limit of the sum is quite tedious, even if $f(x)$ is a very simple function. Both Newton and Leibnitz, more or less at the same time, devised an easy method for evaluating definite integrals. Their method is based upon two celebrated theorems known as **First Fundamental Theorem and Second Fundamental Theorem of Integral Calculus.** These theorems establish the connection between a function and its anti-derivative (if it exists). In fact, the two theorems provide a link between differential calculus and integral calculus. We state below the above important theorems without proofs.

Theorem 9.1 (First Fundamental Theorem of Integral Calculus)

If $f(x)$ be a continuous function defined on a closed interval $[a,b]$ and $F(x)=\int_a^x f(u)du,\ \ a<x<b$ then, $\frac{d}{dx}F(x)=f(x)$. In other words, $F(x)$ is an anti-derivative of $f(x)$.

Theorem 9.2 (Second Fundamental Theorem of Integral Calculus)

If $f(x)$ be a continuous function defined on a closed interval $[a,b]$ and $F(x)$ is an anti-derivative of $f(x)$, then,

$$\int_a^b f(x)dx = F(b)-F(a).$$

Note

Since $F(b)-F(a)$ is the value of the definite integral (Riemann integral) $\int_a^b f(x)dx$, any arbitrary constant added to the anti-derivative $F(x)$ cancels out and hence it is not necessary to add an arbitrary constant to the anti-derivative, when we are evaluating definite integrals. As a short-hand form, we write $F(b)-F(a)=\left[F(x)\right]_a^b$. The value of a definite integral is unique.

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